

Seungk Kang

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Education

Korea Advanced Institute of Science and Technology (KAIST)

Bachelor of Science – Chemistry

Daejeon, Republic of Korea
2022 – Present
(Expected Graduation: early 2027)

Korea Science Academy (KSA)

High School Diploma – Chemistry

Busan, Republic of Korea
2019 – 2021

Skills

Computational Skills

Python (incl. RDKit, NumPy, Pandas); PyTorch; Linux/Bash (HPC/Slurm); Git/GitHub; Machine Learning modeling; DFT calculation (incl. ORCA, ADF); Molecular Dynamics (MD) simulation (Amber); QM/MM simulation; Crystal structure visualization (Mercury, ChemDraw, Chemcraft, Materials Studio, CSD).

Experimental & Analytical Skills

Characterization: Powder X-ray diffraction (PXRD), SEM, EDS, TGA, UV-Vis, IR spectroscopy, Surface area & porosimetry analysis; **Synthesis:** Glove box handling (Inert atmosphere), Ligand purification, TLC, Glass blowing (vacuum line), Electrode fabrication using ITO film.

Research Experiences

Intelligent Chemistry Lab | KAIST

Undergraduate Researcher

Advisor: **Prof. Woo Youn Kim**
Dec. 2025 – present

- Developing a neural free energy estimator (NFEE) for estimating free energy differences of small-molecule ensembles using learned nonequilibrium generative paths.
- Building AD-Agent, a mechanism-aware multi-agent framework for drug repurposing in Alzheimer's disease.
- Studied induced-fit antibody–antigen interaction modeling with a Riemannian Diffusion Bridge-based E(3)-equivariant generative framework.

Baik Group | KAIST

Undergraduate Researcher

Advisor: **Prof. Mu-Hyun Baik**
Jan. 2024 – Dec. 2025

- Studied the mechanism of enantioselective amino-tetralin synthesis catalyzed by an indenyl-Rh complex.
- Performed DFT benchmarking studies on substituent-dependent reactivity in relation to Hammett substituent constants.
- Investigated the mechanism of a solid-state stereoselective [2+2] photocycloaddition.

Soft Material Assembly Group | KAIST

Undergraduate Researcher

Advisor: **Prof. Dong Ki Yoon**
Jun. 2023 – Dec. 2023

- Studied tunable electrical conductivity anisotropy in a MOF–polymer composite using Cu₃(HHTP)₂.
- Investigated the applications of MOF–polymer composites as selective gas sensors.

Choi Group | KSA

Student Researcher

Advisor: **Prof. Eun Young Choi**
Apr. 2019 – Dec. 2021

- Performed gas and ion exchange experiments on zeolite single crystals under high-vacuum conditions.
- Developed protocols to successfully activate fragile porphyrin-based metal–organic frameworks to improve their gas adsorption capacity.

Singapore International Collaborative Research Program

- Synthesized Zn-, Co-, and Ag-tartrate metal–organic frameworks and investigated their antibacterial properties
- Studied biocompatible metal-tartrate frameworks synthesized by the room-temperature rapid self-assembly method

Publications

[1] “Enantioselective Olefin 1,2-Arylation Catalyzed by a Planar Chiral Indenyl-Rhodium(III) Complex” Patrick Gross, Hyoju Choi, Wesley Pullara, Hoyoung Im, Khang Ung, **Seungk Kang**, Mu-Hyun Baik*, Simon B. Blakey*. *ACS Catal.* **2026**, 16(4), 3453-3463.

[2] “On-Demand Tunable Electrical Conductance Anisotropy in a MOF-Polymer Composite.” Taegyung Hong, Changjae Lee, Yeongseo Bak, Geonhyeong Park, Hongju Lee, **Seungk Kang**, Tae-Hyun Bae, Dong Ki Yoon*, Jesse G. Park*. *Small*. **2024**, 20.18: 2309469.

Selected Presentations

Spring Conference of the Korean Chemical Society: Poster presentation | **2025**

Chiral Indenyl Rh-Catalyzed Enantioselective Olefin 1,2-Arylation

Research Presentation for Chemistry Department Workshop: Oral presentation | **2024**

Chiral Indenyl Rh-Catalyzed Enantioselective Olefin 1,2-Arylation

Future Chemists Research Conference of the Korean Chemical Society: Poster presentation | **2021**

Biocompatible MOFs Developed by Facile Ultra-Fast Room Temperature Synthesis

Thailand International Science Fair: Oral presentation | **2021**

Improvement of Gas Adsorption Efficiency of Fragile Porphyrin-based MOFs

Autumn Conference of the Korean Polymer Society: Poster presentation | **2020**

Improvement of Gas Adsorption Efficiency of Fragile Porphyrin-based MOFs

Japan Super Science Fair: Oral presentation | **2020**

Optimizing Synthesis Conditions of PPF-4: Effect of Acid Addition on Crystal Growth

Honors and Awards

Global Entrepreneurship Summer School: *Best participant award* | **2024**

IR pitching contest in Global Entrepreneurship Summer School: *2nd Prize* | **2024**

Talent Award of Korea: *National Prize* | **2021**

Awarded by the Deputy Prime Minister and Minister of Education to 100 individuals nationwide who exemplify future leadership

Meritorious Service Award at the Graduation of the Korean Science Academy | **2021**

Scholarship recipient from the Scholarship Foundation for Korean Science Academy | **2021**

Academic Excellence Award and Comprehensive Studies Award in Chemistry Department, Korean Science Academy (1st and 2nd semesters) | **2021**

Academic Excellence Award and Comprehensive Studies Award in Chemistry Department, Korean Science Academy (1st and 2nd semesters) | **2020**

World Invention Creativity Olympic (WICO): *Gold Prize* | **2017**

Korean Youth Invention Idea Contest by the Korean Ministry of Science and ICT: *The Minister's Award* | **2017**

Youth Science Debate Competition (National Competition): *Silver Prize* | **2015**

Extra Activities

Leader and CEO of mock start-up "KAIST ENablers" | **2024**

Leader of KAIST undergraduate Band "CarpeDiem" | **2022 – 2023**

Leader of Singapore International Cooperative Research Team | **2021**

Leader of Korea Science Academy Band "Loony" | **2019 – 2021**